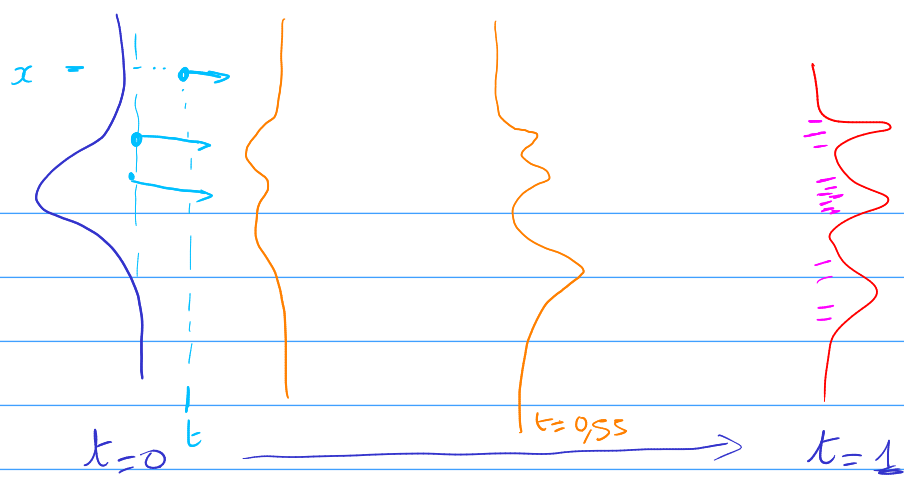




velocity field : $u : (x, t) \mapsto u(x, t) \in \mathbb{R}^d$
 $\xleftarrow{\text{dimensionality}} (\mathbb{R}^d, [0, 1]) \rightarrow \mathbb{R}^d$
 $d = 28 \times 28$

probability path: $\forall t, p_t(x) \cdot (p(r, x))$

CFM loss

$$E_{t \sim U[0,1]} E_{\substack{x_0 \sim \mathcal{N}(\theta, I_d) \\ x_1 \sim \text{Dataset}}} \left[\left\| \mu_\theta(x, t) - \underbrace{(x_1 - x_0)}_v \right\|^2 \right]$$

$x = (1-t)x_0 + tx_1$

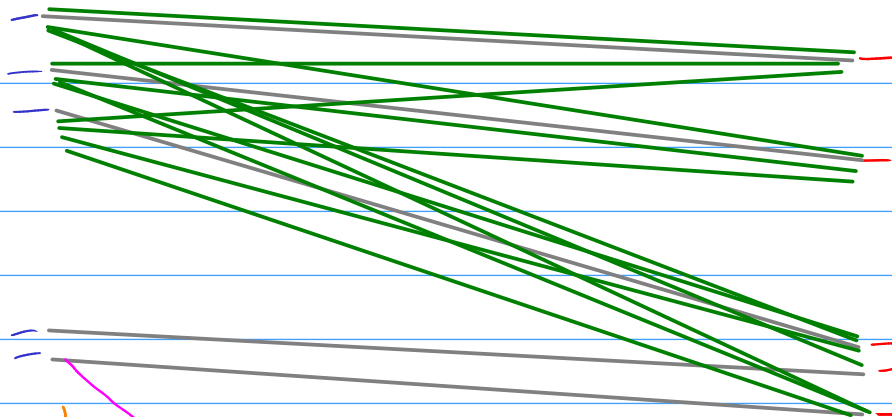
$$(v_1, v_2, v_3, \dots, v_n), \quad \forall_i x_i \in \mathbb{R}$$

$$\underset{\theta}{\operatorname{argmin}} E \left[\|\theta - v_i\|^2 \right] = \underset{\theta}{\operatorname{argmin}} \sum_i (\theta - v_i)^2$$

$$\Rightarrow \theta = \operatorname{mean}(v)$$

Dataset 1

Dataset 2



$$p(x^i | t, \bar{z}) = \mathcal{N}(tx^i; (1-t)^2)$$

